Gruyere Alpha (ATH-969)

Quantitative Equity Market-Neutral Strategy

Track Record: Feb 2015 – Apr 2026 (135 Months / 11.2 Years) | **Benchmark:** S&P 500 Total Return

ATHÉNÉE INVESTMENT

CAPITAL INTRODUCTION & THIRD-PARTY MARKETING

CAGR (Net)	Ann. Volatility	Sharpe Ratio	Max Drawdown	Alpha (Ann.)	Beta & Corr.
10.63%	**5.93%**	**1.74**	**-4.38%**	**10.66%**	**-0.02 | -0.06**
S&P 500: 14.08%	S&P 500: 15.14%	SPX: 0.95 | Sortino: 4.45	SPX: -23.87% | Calmar: 2.43	t-stat: 5.81	Beta t-stat: -0.72

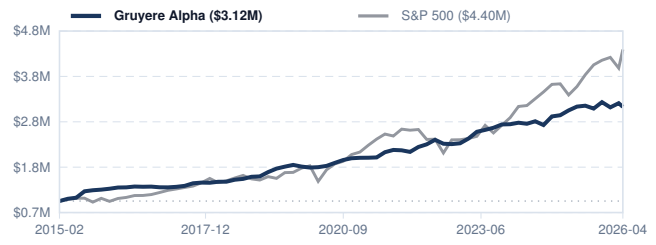
Performance, Risk & Hit Rate Indicators

Indicator	Fund	S&P 500
CAGR (Annualized)	**10.63%**	14.08%
Annualized Mean Return	10.31%	14.38%
Annualized Volatility	5.93%	15.14%
Proxy Sharpe Ratio ($r_f = 0\%$)	**1.74**	0.95
Annualized Sortino Ratio	4.45	1.55
Maximum Drawdown (MDD)	**-4.38%**	-23.87%
Calmar Ratio (CAGR / |MDD|)	2.43	0.59
Cumulative Total Return	211.53%	340.03%
Best / Worst Monthly	+9.08% / -2.87%	+12.82% / -12.35%
Positive Months (Hit Rate)	70.4% (95/135)	69.6% (94/135)
Crisis Hit Rate (Fund > 0 | SPX < 0)	**78.0% (32/41)**	–
Up-Mkt Loss (Fund < 0 | SPX > 0)	33.0% (31/94)	–

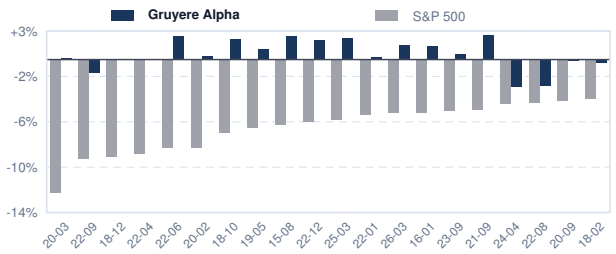
Alpha, Beta, Correlation & Tail Analysis (vs. S&P 500)

Quantitative Metric	Estimate / Diagnostic
Annualized Alpha (α) & t-stat	**10.66% (t=5.81)**
Market Beta (β_{SPX}) & t-stat	-0.02 (t=-0.72) ($R^2 = 0.4\%$)
Pearson Correlation (r) to S&P 500	**-0.06** (95% Blk CI: [-0.19, 0.12])
Spearman Rank Correlation (ρ)	-0.09 (95% Blk CI: [-0.24, 0.11])
Upside / Downside Capture Ratio	Up: **0.21** | Down: **-0.30**
Cond. Up-Market (SPX > 0)	$\alpha = 10.3\%,\ \beta = -0.04$
Cond. Down-Market (SPX < 0)	$\alpha = 21.1\%,\ \beta = 0.15$
Tail Corr. (SPX $< Q_1$, Worst 25%)	$r = 0.26,\ \rho = 0.16\ (n = 34)$
Tail Corr. (SPX $< D_1$, Worst 10%)	$r = 0.54,\ \rho = 0.49\ (n = 14)$
Worst 20% SPX Months Avg. Return	Fund: +0.70% vs SPX: -5.34%
Cross-Asset Beta (WTI / BTC)	WTI β: -0.00 | BTC β: 0.01
Ljung-Box Serial Correlation	$p_1 = 0.06,\ p_3 = 0.21$ (Pass)

Cumulative Growth of $1M USD (Fund vs. S&P 500)



Worst 20% S&P 500 Months



Historical Monthly & Yearly Net Return Matrix (%)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Yearly
2015	-	-0.1%	5.4%	-1.1%	3.4%	9.1%	3.9%	2.2%	-0.2%	0.0%	1.0%	0.4%	**26.2%**
2016	1.2%	2.1%	-0.0%	0.4%	-0.1%	0.9%	0.6%	0.1%	-0.6%	0.1%	0.1%	-0.7%	**4.2%**
2017	-0.4%	0.1%	-0.3%	0.7%	0.3%	0.3%	1.3%	2.3%	1.9%	0.9%	0.2%	-0.3%	**7.3%**
2018	-0.0%	-0.3%	1.9%	-1.2%	1.2%	1.8%	1.3%	0.7%	0.6%	1.9%	1.3%	0.1%	**9.6%**
2019	0.4%	2.1%	4.1%	3.8%	0.9%	1.3%	1.2%	1.0%	1.1%	0.3%	-2.3%	-1.1%	**13.5%**
2020	-0.2%	0.3%	0.1%	1.6%	0.1%	1.3%	2.5%	3.6%	-0.2%	3.1%	-1.0%	1.1%	**12.9%**
2021	-0.5%	-0.2%	0.2%	-0.2%	0.6%	2.8%	3.1%	0.1%	2.2%	-1.4%	1.0%	-1.9%	**5.9%**
2022	0.2%	1.8%	3.4%	-0.0%	2.7%	2.2%	2.2%	-2.4%	-1.3%	-0.7%	0.3%	1.8%	**10.5%**
2023	-0.9%	1.1%	3.4%	4.7%	1.6%	2.6%	-0.9%	1.3%	0.5%	0.7%	2.2%	-0.9%	**16.3%**
2024	1.1%	-1.2%	2.4%	-2.5%	1.7%	0.6%	1.4%	-0.2%	-2.9%	3.9%	3.1%	0.3%	**7.8%**
2025	0.7%	1.7%	2.0%	1.5%	1.3%	-0.0%	0.7%	-0.7%	-1.4%	3.1%	1.5%	-1.1%	**9.6%**
2026	-2.6%	1.7%	1.3%	-2.5%	-	-	-	-	-	-	-	-	**-2.2%**

Fund Terms & Commercial Overview

Vehicle:	**Commingled Fund**	Return Basis:	**Net of Fees**	Fee Structure:	**2% Mgmt / 20% Perf**	Liquidity:	**Quarter Liquidity**
Lock-Up Period:	**No Lockup**	Min. Investment:	**1M USD**	Current AUM:	**250M**	Domicile:	**Cayman Islands**

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Capital Introduction & Third-Party Marketing Memorandum. Strictly for Institutional & Qualified Investors. Please see Page 2 for Risk & Legal Disclosures.

Risk Disclosures, Legal Notice & Quantitative Appendix

Fund Reference: Gruyere Alpha (ATH-969) (Feb 2015 - Apr 2026 (135 Months / 11.2 Years))

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- **Model, Regime Shift & Tail-Correlation Risk:** Quantitative and systematic strategies rely on mathematical models and historical statistical relationships (such as low equity beta or negative downside capture) that may break down or experience sudden correlation spikes during systemic liquidity crises.
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Appendix A: Extended Statistical, Bootstrap & Factor Diagnostics

Diagnostic Family	Methodology / Statistical Specification	Empirical Value
Pearson Correlation CI	95% Confidence Interval (IID Bootstrap vs. Moving Block $B = 3$, 1,000 iters)	Std: [-0.21, 0.10] | Blk: [-0.19, 0.12]
Spearman Rank Corr. CI	95% Confidence Interval (Rank Monotonic Co-Movement, IID vs. Block $B = 3$)	Std: [-0.25, 0.09] | Blk: [-0.24, 0.11]
Worst-Case Co-Dependence	97.5th Percentile Upper-Bound Block Bootstrap Correlation (PSD Projection)	$\rho_{\text{wc}} = 0.12$
Serial Correlation (Lag 1 & 3)	Ljung-Box Q-Statistic p-values testing for return smoothing / illiquidity bias	$p_1 = 0.06,\ p_3 = 0.21$ (Pass)
Conditional Up-Market OLS	Annualized Alpha & Beta estimated during positive S&P 500 months ($n = 94$)	$\alpha = 10.3\%,\ \beta = -0.04\ (t_\alpha = 3.23)$
Conditional Down-Market OLS	Annualized Alpha & Beta estimated during negative S&P 500 months ($n = 41$)	$\alpha = 21.1\%,\ \beta = 0.15\ (t_\alpha = 3.52)$
Commodity Factor (WTI)	OLS Regression against Front-Month WTI Crude Oil Monthly Returns	$\beta = -0.00\ (t = -0.35),\ \alpha = 9.9\%$
Digital Asset Factor (BTC)	OLS Regression against Bitcoin (BTC-USD) Monthly Returns	$\beta = 0.01\ (t = 0.94),\ \alpha = 9.2\%$

Appendix B: Quantitative Framework & Indicator Definitions

- **CAGR & Annualized Volatility:** Compound Annual Growth Rate is computed geometrically as CAGR $= \left(\prod_{t=1}^{T}(1 + r_t)\right)^{12/T} - 1$. Annualized Volatility is sample standard deviation scaled by $\sqrt{12}$: $\sigma_{\text{ann}} = \hat{\sigma}_m\sqrt{12}$. Proxy Sharpe Ratio assumes $r_f = 0\%$ ($\bar{r}_{\text{ann}}/\sigma_{\text{ann}}$).
- **Maximum Drawdown (MDD) & Calmar Ratio:** Maximum peak-to-trough decline of cumulative compounded wealth $W_t = \prod_{s\le t}(1 + r_s)$, defined as MDD $= \min_t\left((W_t - \max_{s\le t} W_s)/\max_{s\le t} W_s\right)$. Calmar Ratio is CAGR/|MDD|.
- **OLS Alpha (α), Beta (β) & Capture Ratios:** Estimated via Ordinary Least Squares $r_{\text{fund},t} = \alpha_m + \beta r_{\text{SPX},t} + \varepsilon_t$, where $\alpha_{\text{ann}} = 12 \times \alpha_m$. Upside (Downside) Capture measures the ratio of the fund's mean return to the S&P 500's mean return during positive (negative) S&P 500 months.
- **Tail Correlations & Moving Block Bootstrap:** Tail correlations evaluate linear (r) and rank (ρ) co-movement during S&P 500 drawdowns below the 25th percentile (Q_1) and 10th percentile (D_1). Moving Block Bootstrap ($B = 3$ months) preserves local serial dependencies when constructing 95% confidence intervals.

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